10.6 Circumference and arc length

EASY

Learning intention We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use $C = \pi d$ or $C = 2\pi r$ to find a circumference
- ☐ I can find an arc length: $\theta \div 360 \times 2\pi r$
- ☐ I can find the perimeter of sectors and composite shapes

NOTES Copy what your teacher writes on the board. **Date** _____

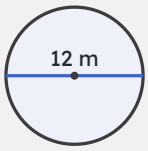
WE DO With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

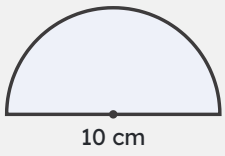
a Circumference



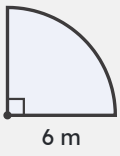
b Circumference



c Perimeter



d Perimeter



YOU DO On your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER $d = 2r \cdot r = d \div 2 \cdot C = \pi d \cdot$ Use the π key, then round to one decimal place.

ROUND 1 Questions 1–4: find the diameter or the radius.

1 r = 4 cm. Find d.
Formula: d = 2r
Substitute: _____
Answer: _____

2 d = 10 m. Find r.
Formula: r = d ÷ 2
Substitute: _____
Answer: _____

3 r = 6.5 cm. Find d.
Formula: d = 2r
Substitute: _____
Answer: _____

4 d = 9 m. Find r.
Formula: r = d ÷ 2
Substitute: _____
Answer: _____

ROUND 2 Questions 5–8: the diameter is given. Find the circumference.

5 d = 6 cm
Formula: C = πd
Substitute: _____
Answer: _____

6 d = 10 m
Formula: C = πd
Substitute: _____
Answer: _____

7 d = 2.5 m
Formula: C = πd
Substitute: _____
Answer: _____

8 d = 14 cm
Formula: C = πd
Substitute: _____
Answer: _____

YOU DO On your own: keep going, one round at a time. Show your working. 15 QUESTIONS

REMEMBER C = 2πr = πd · Semicircle P = ½πd + d · Quadrant P = ¼ × 2πr + 2r · Find d first when r is given.

ROUND 3 Questions 9–11: the radius is given. Find the circumference.

9 r = 2 cm

Formula C = 2πr
Substitute
Answer

10 r = 8 m

Formula C = 2πr
Substitute
Answer

11 r = 1.5 m

Formula C = 2πr
Substitute
Answer

ROUND 4 Questions 12–14: is it r or d? Write the right formula, then find the circumference.

12 r = 4 cm

Formula
Substitute
Answer

13 d = 9 m

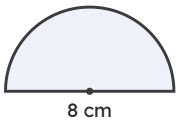
Formula
Substitute
Answer

14 r = 12 mm

Formula
Substitute
Answer

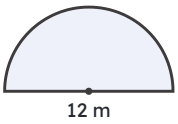
ROUND 5 Questions 15–17: find the perimeter of each semicircle.

15



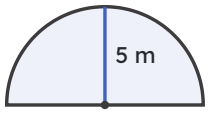
Formula P = ½πd + d
Substitute
Answer

16



Formula P = ½πd + d
Substitute
Answer

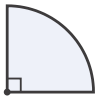
17



Formula P = ½πd + d
Substitute
Answer

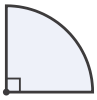
ROUND 6 Questions 18–20: find the perimeter of each quadrant.

18



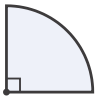
Formula P = ¼ × 2πr + 2r
Substitute
Answer

19



Formula P = ¼ × 2πr + 2r
Substitute
Answer

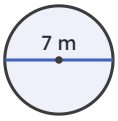
20



Formula P = ¼ × 2πr + 2r
Substitute
Answer

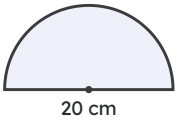
ROUND 7 Questions 21–23: circle, semicircle or quadrant? Write the formula, then find the perimeter.

21



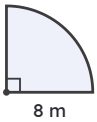
Formula
Substitute
Answer

22



Formula
Substitute
Answer

23



Formula
Substitute
Answer

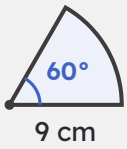
10.6 Circumference and arc length

MEDIUM

WE DO With your teacher: copy what they write on the board.

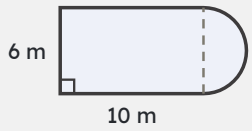
Example 2 **Give each answer correct to one decimal place.**

a Arc length



b Perimeter of the sector in part a

c Perimeter



d The radius of a circle with C = 50 cm

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

REMEMBER Arc = θ ÷ 360 × 2πr (θ = angle at the centre) · Sector P = arc + 2r · Composite: add the outside edges only

1 Find the arc length of a 120° sector with r = 3 m.

Formula _____
Substitute _____
Answer _____

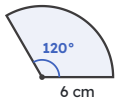
2 A kite has diagonals of 6 m and 5 m. Find its area. 10.5

Formula _____
Substitute _____
Answer _____

3 Find the arc length of a 45° sector with r = 10 cm.

Formula _____
Substitute _____
Answer _____

4 Perimeter



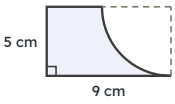
Formula _____
Substitute _____
Answer _____

5 Perimeter



Formula _____
Substitute _____
Answer _____

6 Perimeter. Cut-out: r = 5 cm



Formula _____
Substitute _____
Answer _____

7 Write the limits of accuracy of 4.5 m. 10.2

_____

8 A circle has a circumference of 18.85 m. Find its radius.

Formula _____
Substitute _____
Answer _____

9 Percentage error of 20 cm, to the nearest cm? 10.3

Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A round garden bed has a diameter of 3.2 m. Edging costs $12 per metre. Find the cost to edge the bed, to the nearest dollar.

Answer _____

2 A bike wheel has a radius of 33 cm. How many full turns does it make to travel 1 km?

Answer _____

3 A running track is a rectangle 84 m long with a semicircle on each end. Each semicircle has a diameter of 64 m. Find one lap, to the nearest metre.

Answer _____